
S20196 - H.M.S.V.Jayasekara

CSC4093

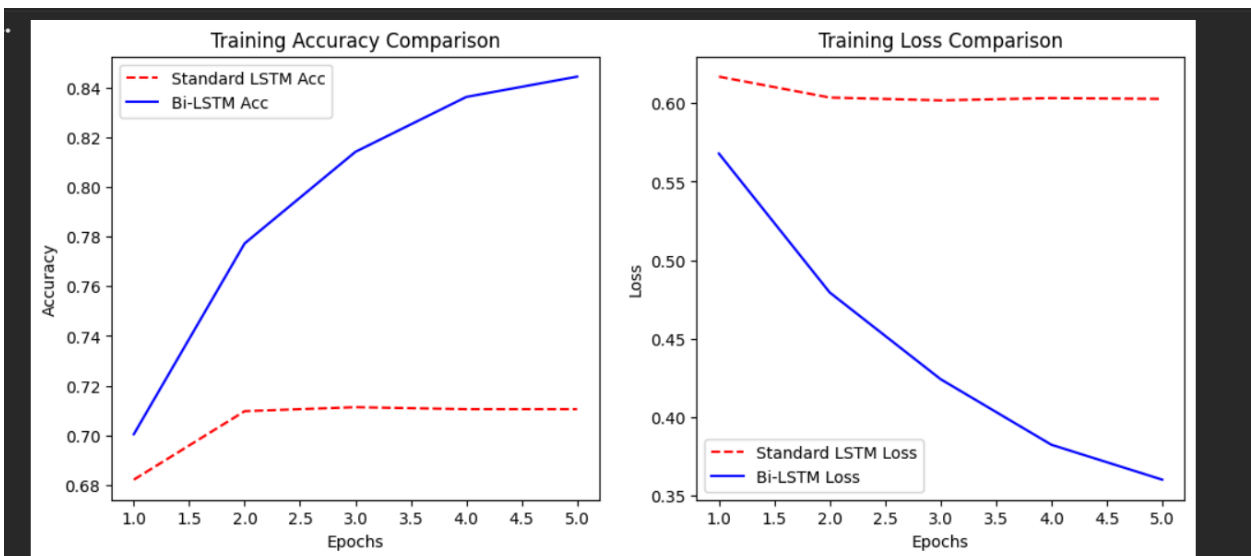
**Neural Networks and Deep Learning
Programming Assignment 01 : LSTMs**

**Report of
Sentiment Analysis of Personal Health
Mentions using LSTM and Bi-LSTM**

Performance Comparison

The following table and plots summarize the performance of the Standard LSTM and the Bidirectional LSTM (Bi-LSTM) after 5 epochs of training on the tweet dataset.

--- Final Model Performance Comparison ---					
	Model Architecture	Test Accuracy (%)	Training Loss (Final)	Total Parameters	Accuracy Improvement
0	Standard LSTM	71.49	0.6027	354929	-
1	Bidirectional LSTM	81.78	0.3604	389745	+10.29%



Accuracy Plot: Shows the Bi-LSTM (blue line) consistently climbing toward 84% training accuracy, while the Standard LSTM (red dashed line) plateaus early around 71%.

Loss Plot: Shows the Bi-LSTM loss sharply decreasing, indicating better convergence, whereas the Standard LSTM loss remains high and relatively flat.

Discussion on Performance

i. Comparison of Results

The experimental results show a significant performance lift when using a Bidirectional architecture. The Bi-LSTM achieved an accuracy of 81.78%, outperforming the Standard LSTM's 71.49% by more than 10%.

Looking at the training history, the Standard LSTM struggled to learn complex patterns in the data, as shown by its high loss (~0.60) and stagnant accuracy. In comparison, the Bi-LSTM followed a much smoother and more stable learning pattern. Its loss consistently decreased with each epoch, showing that the model was actually learning and improving over time instead of struggling.

Although the Bi-LSTM includes slightly more parameters (around 35,000 extra), the additional computational cost is relatively small.

The Bi-LSTM outperformed the standard LSTM mainly because of how it understands sequences.

ii. Why the Bi-LSTM Performed Better

The better performance of the Bi-LSTM in this tweet classification task can be explained by the way it processes sequential data

Dual contextual understanding:

A regular LSTM reads text in one direction, usually from left to right. But in sentiment analysis, meaning often depends on what comes next. Take the phrase "not very happy." The real sentiment becomes clear only after seeing the full phrase. A Bi-LSTM reads the sentence in both directions at the same time, so each word is interpreted using both previous and upcoming context. That fuller view helps the model make smarter decisions.

Better feature extraction for short texts:

Tweets are short, messy, and often informal. Since the Bi-LSTM processes text forward and backward, it effectively gathers more contextual signals for each word. This makes it stronger at extracting meaningful patterns from limited text, which is very important in tweet classification.

Improved handling of long-term dependencies:

By adding a backward pass, the model shortens the distance that information needs to travel within the network. It reduces issues like vanishing gradients better than a one directional LSTM.

iii. Classification Insights

The classification report shows that the Bi-LSTM performs especially well at identifying negative tweets, achieving 83% precision and 93% recall. This suggests that negative sentiment in the dataset is often expressed more directly, making it easier for the model to detect.

However, the model is slightly less precise when identifying positive tweets (75% precision). This may indicate that positive sentiment in the dataset is expressed in more subtle, mixed, or even sarcastic ways, making it harder for the model to distinguish compared to negative tweets, which often is direct, clear and with strong wording. As a result, the model may find it more challenging to clearly separate positive tweets.

Detailed Bi-LSTM Performance:

	precision	recall	f1-score	support
Negative	0.83	0.93	0.88	2364
Positive	0.75	0.54	0.63	967
accuracy			0.81	3331
macro avg	0.79	0.73	0.75	3331
weighted avg	0.81	0.81	0.80	3331